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ROTATIONAL FORCE ASSISTANCE FOR CROSSBOW DRAWING

Abstract

A rotational force assistance for crossbow drawing device and method for use with a variety of crossbows having a variety of bowstring drawing mechanisms having rotational input units, providing a unit case, an electric powered drive unit, a charge-storage unit battery or capacitor, a charging port, and a controller, where operation of a draw activator causes the drive unit to rotate, turning a rotational-input-unit adapter that matches the crossbow's rotational input unit, which transfers rotational force from the drive unit through the rotational-input-unit adapter to the rotational input unit of the crossbow, causing the crossbow's bowstring drawing mechanism to draw and lock the bowstring. Preferred embodiments additionally provide a grip surface on the unit case, an indicator, such as a light to indicate charging and operational status, a tether point providing for attachment of a tether, such as a strap, lanyard, or clip, and a bracing extension providing for bracing against the crossbow structure. A releasing embodiment additionally provides rotational force assistance for controlled release of tension on the bowstring, where operation of a release activator causes the drive unit, rotational-input-adapter, and the crossbow's rotational input unit to rotate in an opposite direction, thereby releasing tension on the bowstring in a controlled and safe manner.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] This invention provides a rotational force assistance for crossbow drawing device and method for use with a variety of crossbows having a variety of bowstring drawing mechanisms having rotational input units.

[0002] Crossbows have been known and used by many groups of people over many centuries. A tiller or stock provides the frame of the weapon. A prod or lath is mounted on the stock, and a bowstring or sinew is mounted across the lath. In use, the bowstring is pulled back or drawn and is held in the drawn position by a locking mechanism. The prod or lath is pulled into a bent position, causing a large amount of elastic potential energy to be stored in the bent lath and the bowstring. A bolt, quarrel, or arrow is placed on the stock, ahead of the bowstring. When a trigger is pulled and the lock is released the energy in the bent lath and the bowstring is transformed into kinetic energy, and the bolt is shot from the crossbow with great force.

[0003] Compound crossbows have been developed, and recurve crossbows have been improved, so that modern crossbows can store and release a great amount of energy. But a great amount of stored and released energy requires a great amount of energy input to draw and lock the bowstring, and therefore requires some kind of device or mechanism to provide a mechanical advantage so that a person's input of 5 to 50 pounds of energy can be transformed into the 175 pounds and more required to draw the bowstring of a typical modern crossbow.

[0004] The oldest and most basic method for drawing and locking the bowstring is to point the crossbow at the ground and step into a stirrup fixed to the end of the stock, and to directly pull the bowstring up and into the lock. This requires a great amount of strength and effort. Many devices or tools have been used with or incorporated into crossbows in order to achieve greater energy storage and release and to reduce the amount of manual exertion required to draw and lock the bowstring. Devices or tools or spanning mechanisms such as winch or cord pulleys, gaffe levers, cranequin rack-and-pinion mechanisms, ratchets, screws, and windlass-like mechanisms have been used.

[0005] Many modern crossbows can have bowstring drawing mechanisms either incorporated into the crossbow or available for retrofit into the crossbow. Such modern bowstring drawing mechanisms have a rotational input unit in the form of a socket or a stud or similar rotating structure that allows the attachment of a hand crank. Turning the hand crank with 5 to 50 pounds of force is transformed into the much larger force required to draw and lock the bowstring. Multiple turns of the rotational input unit at lower force are transformed into a shorter movement of the bowstring at greater force. A ratchet or pawl mechanism can divide the bowstring drawing force into discrete steps, avoiding the circumstance of having to manually sustain the entire total drawing force as with a longbow. Generally, where a bowstring drawing mechanism is designed to require only a small force of about 5 pounds, many more revolutions of the rotational input unit will be required to produce the bowstring drawing force than would be required where a greater force of about 50 pounds is used. The manual crank handle, which is attached to the rotational input unit, is usually long in order to increase leverage, and the user might be required to turn the crank handle in a large circle requiring significant extension and movement of the hand and arm.

[0006] Even though the bowstring drawing mechanisms of modern crossbows significantly reduce the amount of user effort required to draw and lock the bowstring, the reduced effort still might be too much for some users having limitations of strength, mobility, or endurance. A device for rotating the rotational input units of a variety of crossbows, at optimum speeds and with optimum

torque, is needed.

[0007] A common and constant problem with using a crossbow is releasing the tension on the drawn and locked bowstring before putting the crossbow away. Usually, the loaded bolt must be shot into the ground or into a target, potentially damaging the bolt itself or damaging something or someone with the bolt. The crossbow cannot be dry fired without the bolt because doing so is likely to damage both the crossbow and persons near the crossbow. Some modern crossbows have a releasing or de-cocking feature where rotation of the rotational input unit in the direction opposite the direction for tightening will cause the tension on the drawn bowstring to be released in a gradual and controlled way. A user performing such a releasing operation is required to make the same movements, in reverse, and exert substantially the same amount of force, as the bowstring drawing operation. And a user having limitations of strength, mobility, or endurance will have substantially the same problems performing the releasing operation as performing the drawing operation. A device additionally providing for the releasing operation is also needed.

[0008] U.S. Pat. No. 7,836,871 for a “Powerstroke Crossbow,” issued on Nov. 23, 2010 to inventor James J. Kempf, provides for a crossbow having an increased powerstroke and reduced noise. The powerstroke is increased by inverting the limb orientation from the standard crossbow arrangement and locating string guides at least partially forward and rearward of the ends of the limbs. The bowstring is drawn from the tops of the string guides to maximize the powerstroke, reducing noise and increasing the retained and delivered energy over existing crossbows.

[0009] U.S. Pat. No. 9,341,432 for a “Drawstring Cocking Assembly,” issued on May 17, 2016 to inventor Clem Wohleb, provides for a cocking assembly including a housing structured to have a rod portion and a gearbox portion. The rod portion is aligned with a drawstring of a weapon. A rod is rotatably positioned within the interior of the rod portion. A motor is coupled to the gearbox portion. The motor rotates in a first direction and a second direction. An actuator actuates the motor in the first and second directions. Each of a plurality of gears is rotatably coupled to an associated one of the motor and the rod. Thus, the plurality of gears is enmeshed to transfer a rotational torque of the motor into the rod. A yoke is structured to have a block and a saddle. The block threadably engages the rod such that the yoke is urged forwardly and rearwardly along the rod. Thus, the yoke cocks the drawstring of the weapon.

[0010] U.S. Pat. No. 6,095,128 for “Crossbow Bowstring Drawing Mechanisms,” issued on Aug. 1, 2000 to assignee Tenpoint Crossbow Technologies, provides for crossbow bowstring drawing mechanisms that are integrated into or secured in the crossbow handle and which provide a straight and balanced draw to the crossbow bowstring to cock the crossbow bowstring in position in the crossbow trigger mechanism ready for firing. The cocking system can be manually operated or motorized, and can be manufactured as part of a crossbow or retrofit into a crossbow. An internal or external source of rotational power, such as hand crank, power screwdriver or an electrical motor, is preferably utilized in conjunction with a claw member including a bowstring engaging portion to translate rotation of the input source to longitudinal movement of the claw member to draw or release the crossbow bowstring resulting with minimal effort being expended by the user.

[0011] U.S. Pat. No. 8,375,928 for a “Slip Clutch,” issued on Feb. 19, 2013 to assignee Hunter's Manufacturing Company, Inc., provides for a bowstring drawing mechanism for drawing and controllably releasing a crossbow bowstring that comprises a clutch mechanism. The clutch mechanism protects the bowstring drawing mechanism from damage caused by over-cranking and allows the user to controllably release the bowstring from the drawn position.

[0012] U.S. Pat. No. 5,220,906 for a “Device to Draw the Bowstring of a Crossbow,” issued on Jun. 22, 1993 to assignee Horton Manufacturing Company Inc., provides for a device for cocking a crossbow having a bow and a bowstring including a main housing, a movable shaft, a device operatively connected to the movable shaft to laterally move the movable shaft, and a back element mounted on the movable shaft to engage the bowstring, such that when the movable shaft moves, the bowstring is drawn.

[0013] U.S. Pat. No. 4,478,202 for a “Force Multiplying Archery Bow,” issued on Oct. 23, 1984 to inventor Jeffrey J. Anderson, provides for a archery bow with a system whereby of which the force applied to bending the bow is multiplied when the drawstring is pulled back by the user. This is accomplished by attaching a second string between the bow limbs and using a mechanical gearing or other force-multiplying system actuated by a lever which is pulled with the main drawstring. At full draw both strings are simultaneously released. Electrical, hydraulic or pneumatic force multiplying devices can also be used to draw the second string.

[0014] U.S. Pat. No. 10,408,558 for a “Crossbow Having an Energizer,” issued on Sep. 10, 2019 to assignee Bakke Invest AS, provides for A crossbow having an energizer. The crossbow, in an embodiment, includes a stock, a body coupled to the stock, and a plurality of limbs. Each of the limbs has a coupled limb end and an uncoupled limb end. The energizer, which includes an electrical power source, is operatively coupled to the limbs.

[0015] U.S. Pat. No. 10,458,743 for a “Crossbow with Built In Cocking Device,” issued on Oct. 29, 2019 to inventors James J. Kempf et al., provides for a crossbow with built in cocking device including a manual crank and a built-in, removable motor gearbox assembly and power source. Rotational forces applied to a drive shaft rotate in a first direction, causing first and second string carriages to engage a bow string, moving the bow string towards a string catch. Once the string catch is latched to the bow string, a drive shaft is rotated in a second direction to move first and second string carriages to a rest position

SUMMARY OF THE INVENTION

[0016] This invention provides a rotational force assistance for crossbow drawing device and method for use with a variety of crossbows having a variety of bowstring drawing mechanisms having rotational input units.

[0017] The rotational force assistance for crossbow drawing device provides a unit case, an electric powered drive unit, a charge-storage unit battery or capacitor, a charging port, and a controller, where operation of a draw activator causes the drive unit to rotate, turning a rotational-input-unit adapter that matches the crossbow's rotational input unit, which transfers rotational force from the drive unit through the rotational-input-unit adapter to the rotational input unit of the crossbow, causing the crossbow's bowstring drawing mechanism to draw and lock the bowstring.

[0018] Preferred embodiments of the rotational force assistance for crossbow drawing device additionally provide a grip surface on the unit case, an indicator such as a light to indicate charging and operational status, a tether point providing for attachment of a tether, such as a strap, lanyard, or clip, and a bracing extension providing for bracing against the crossbow structure.

[0019] A releasing embodiment of the rotational force assistance for crossbow drawing device additionally provides rotational force assistance for controlled release of tension on the bowstring, where operation of a release activator causes the drive unit, rotational-input-adapter, and the crossbow's rotational input unit to rotate in an opposite direction, thereby releasing tension on the bowstring in a controlled and safe manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] Reference will now be made to the drawings, wherein like parts are designated by like numerals, and wherein:

[0021] FIG. 1 is a perspective view of the rotational force assistance for crossbow drawing device of the invention in preparation for use;

[0022] FIG. 2 is a perspective view of the rotational force assistance for crossbow drawing device of the invention with the bracing extension extended;

[0023] FIG. 3 is a perspective view of the rotational force assistance for crossbow drawing device

of the invention with the bracing extension retracted;

[0024] FIG. 4 is a perspective view of various embodiments of the rotational-input-unit adapter of the rotational force assistance for crossbow drawing device of the invention;

[0025] FIG. 5 is a perspective view of the rotational force assistance for crossbow drawing device of the invention in use prior to mounting to the crossbow;

[0026] FIG. 6 is a perspective view of the rotational force assistance for crossbow drawing device of the invention in use with the bracing extension extended prior to mounting to the crossbow;

[0027] FIG. 7 is a perspective view of the rotational force assistance for crossbow drawing device of the invention in use mounted to the crossbow;

[0028] FIG. 8 is a perspective view of the rotational force assistance for crossbow drawing device of the invention in use prior to dismounting from the crossbow; and

[0029] FIG. 9 is a perspective view of the rotational force assistance for crossbow drawing device of the invention in use after dismounting from the crossbow.

DETAILED DESCRIPTION OF THE INVENTION

[0030] Referring to FIG. 1, the rotational force assistance for crossbow drawing device **10** and method is intended for use with a variety of crossbows having a variety of bowstring drawing mechanisms having rotational input units. The rotational force assistance for crossbow drawing device **10** is useful especially for users who have difficulty or otherwise do not wish to use the hand cranks supplied for operation of a crossbow's bowstring drawing mechanism. Different brands and models of crossbows have different configurations, sizes, and shapes of the rotational input unit of the bowstring drawing mechanism. For example, the rotational input unit might have a protruding stud, pin, or key, might instead have a socket of square or hexagonal shape, or might have a shaped stud intended to fit into a corresponding socket. Detachable hand-crank handles are also provided with the crossbows, where the crank handle is configured with a mounting connector corresponding to the configuration of the rotational input unit. The rotational force assistance for crossbow drawing device **10** provides rotational-input-unit adapters, which on one end mount to the drive unit **3** of the device, and on the other end conform to and mount to the specific configuration of the rotational input unit of the bowstring drawing mechanism of the specific crossbow, as treated in more detail below. A single unit of the rotational force assistance for crossbow drawing device **10** can be used with different brands and models of crossbows by interchanging the appropriate rotational-input-unit adapters **30**.

[0031] Referring additionally to FIG. 2 and FIG. 3, the rotational force assistance for crossbow drawing device **10** provides a unit case **1** of a size and shape that is easily carried and easily manipulated with one hand. In a preferred embodiment, a grip surface **2** such as a rubber coating or a textured surface or a change in size is provided, as shown. The configuration of the unit body **1** and grip surface **2** improve the user's ability to hold and manipulate the device, including when wearing gloves or in difficult weather or other conditions, such as when moving on foot or in a vehicle, close quarters, or when needing to minimize body movements to avoid detection. The unit case **1** can be made of a hard plastic material. In use, the configuration of the unit case **1** and the intended position for gripping the device provide the user with some mechanical advantage or leverage for resisting any counter-rotational force encountered during operation. Typical dimensions for the illustrated embodiments are approximately 7 to 8 inches on the longest axis, and approximately 1.5 and 2 inches on the sides.

[0032] The unit case **1** encloses a drive unit **3** that operates on electric power, a charge-storage unit **4**, and a controller **5**, treated in detail below. A charging port **6** is provided on the unit case **1**, for the purpose of recharging the charge-storage unit **4**. Optionally, an indicator **7** can be provided for the purpose of showing status information. The indicator **7** can be one or more LED lights.

Optionally, a tether point **11** can be provided for the purpose of providing a mounting point for a tether, such as a strap, clip, lanyard, or other tether for securing the rotational force assistance for crossbow drawing device either to the user's person, the user's clothing or equipment, or to the

crossbow itself.

[0033] A draw activator **8** is provided on the unit case **1** for the purpose of activating the drive unit **3** and mounted rotational-input-unit adapter **30** to apply rotational force to the rotational input unit of the bowstring drawing mechanism of the crossbow in a tightening or bowstring-drawing direction, which is typically clockwise, as treated in detail below.

[0034] Optionally in a bowstring-tension-releasing embodiment, a release activator **9** is also provided on the unit case **1** for the purpose of activating the drive unit **3** and mounted rotational-input-unit adapter **30** to apply rotational force to the rotational input unit of the bowstring drawing mechanism of the crossbow in a releasing or de-cocking direction, which is typically counterclockwise for crossbows having the tension-releasing feature.

[0035] Operation of the bowstring drawing mechanism of a crossbow will either require a larger number of revolutions at a lesser force or a lesser number of revolutions at a greater force. A counter-rotational force will operate upon the device and will correspond to the rotational force applied by the device. The configuration of the unit case **1** provides some mechanical advantage or leverage to help the user resist such counter-rotational force. In a preferred embodiment, a bracing extension **12** is also provided for the purpose of bracing against the crossbow structure to resist the counter-rotational force. The bracing extension **12** protrudes from the unit case **1** toward the crossbow structure, such as the stock of the crossbow. In use, the bracing extension can be allowed to come into contact with and brace against the crossbow structure, which will resist the counter-rotational force so that the user is not required to resist the counter-rotational force. Optionally, the bracing extension **12** can be configured to be a retractable bracing extension, as shown.

[0036] The drive unit **3** operates on DC electric power and transforms electric power into rotational force that is transferred to the rotational input unit of the bowstring drawing mechanism of the crossbow through the appropriate rotational-input-unit adapter **30** corresponding to the specific configuration of the rotational input unit, as treated in detail herein. The output rotational force should provide adequate torque force at relatively low RPMs, corresponding to normal manual operation of the rotational input unit using a hand crank. An electric motor can be used. Where a stepper motor type of electric motor is used there should not be a need to provide additional gearing, ratcheting, or braking components in the drive unit **3** because a stepper motor of the proper size and configuration can directly provide adequate torque force at relatively low RPMs. In embodiments providing bowstring tension releasing capabilities, a stepper motor can provide reversed rotation without the need for reverse gearing. Where a conventional brushed DC electric motor is used, the drive unit **3** can provide additional gearing, ratcheting, braking, or reversing components as necessary and as known in the art.

[0037] The charge-storage unit **4** stores the electric power used by the drive unit **3**. The charge-storage unit **4** can be a battery or group of batteries, or can be a capacitor or supercapacitor, or can be a combination of battery and capacitor technology. As is known in the art, a battery generally holds a greater amount of charge but can only provide usable current at a relatively slow rate, while a capacitor generally holds a smaller total amount of charge but can provide more usable current relatively quickly.

[0038] The charging port **6** provides for charging or recharging the charge-storage unit **4** with electric power from an external source. In a preferred embodiment, the charging port **6** is a USB type of port that allows charging under standard protocols using standard and readily available charging equipment and cables.

[0039] The controller **5** controls the operation of the rotational force assistance for crossbow drawing device **10**, including the charging and recharging of the charge-storage unit **4** through the charging port **6**, the operation of the drive unit **3** when the draw activator **8** is engaged by the user, and where appropriate the operation of the drive unit **3** when the release activator **9** is engaged by the user. The controller **5** can be implemented in electronic circuitry as known in the art.

[0040] The draw activator **8** is a button or switch that is engaged by the user in order to cause the

drive unit **3** with the mounted rotational-input-unit adapter **30** to apply rotational force to the rotational input unit of the bowstring drawing mechanism of the crossbow in a tightening direction, typically clockwise, under the control of the controller **5**.

[0041] In embodiments providing bowstring tension releasing capabilities, the release activator **9** is a button or switch that is engaged by the user in order to cause the drive unit **3** with the mounted rotational-input-unit adapter **30** to apply rotational force to the rotational input unit of the bowstring drawing mechanism of the crossbow in a releasing direction, typically counter-clockwise, under the control of the controller **5**.

[0042] Referring to FIG. **4**, a variety of embodiments of the rotational-input-unit adapter **30** are provided, each of which adapts the drive unit **3** of the rotational force assistance for crossbow drawing device **10** to the specific configuration, size, and shape of the rotational input unit of a specific model or brand of crossbow. The illustrated exemplary embodiments of rotational-input-unit adapters **30** are a keyed rotational-input unit adapter **31**, a square-socket rotational-input unit adapter **32**, a hex-socket rotational-input unit adapter **33**, and a five-sided rotational-input unit adapter **34**. A single unit of the rotational force assistance for crossbow drawing device **10** can be used with a large number of different crossbows by interchanging the appropriate rotational-input-unit adapters **30**.

[0043] Each embodiment of the rotational-input-unit adapter **30** has a crossbow-facing end that is configured to fit the bowstring drawing rotational input unit of the crossbow, and has an opposite drive-unit facing end that is configured to couple with the drive unit **3** by a single, universal, removable, and interchangeable means. The rotational-input-unit adapters **30** can be securely and removably mounted on the drive unit **3** using such means as a magnetic coupling, a spring-loaded ball or pin, or a retaining ring or sleeve.

[0044] Referring to FIG. **5**, in use, initially, an appropriate embodiment of the rotational-input-unit adapter **30** is mounted to the rotational force assistance for crossbow drawing device **10**. The appropriate embodiment of the rotational-input-unit adapter **30** is the embodiment that corresponds to the rotational input unit of the bowstring drawing mechanism of a specific crossbow.

[0045] Referring to FIG. **6**, in embodiments providing a retractable bracing extension **12**, the bracing extension **12** is extended to protrude from the unit case **1** toward the crossbow structure.

[0046] Referring to FIG. **7**, the rotational-input-unit adapter **30**, which has already been mounted on one end to the drive unit **3**, is mounted at the opposite end to the rotational input unit of the bowstring drawing mechanism of the crossbow. For preferred embodiments having a bracing extension **12**, the bracing extension **12** can be placed in contact with the crossbow structure.

[0047] Referring to FIG. **8**, the user engages the draw activator **8**. Under the control of the controller **5** and using electric power from the charge-storage unit **4**, the drive unit **3** transforms electric power into rotational force and applies the rotational force in a tightening direction, through the mounted rotational-input-unit adapter **30**, to the rotational input unit of the bowstring drawing mechanism of the crossbow, causing the bowstring to be drawn and locked.

[0048] Referring to FIG. **9**, after use, the rotational force assistance for crossbow drawing device **10** can be unmounted from the crossbow and can be placed in a pocket or be otherwise put away until needed again.

[0049] For embodiments providing bowstring tension releasing capabilities, in use for releasing bowstring tension, the rotational force assistance for crossbow drawing device **10** with a rotational-input-unit adapter **30** mounted is mounted to the rotational input unit of the bowstring drawing mechanism of the crossbow, and the user engages the release activator **9**. Under the control of the controller **5** and using electric power from the charge-storage unit **4**, the drive unit **3** transforms electric power into rotational force and applies the rotational force in a releasing direction, through the mounted rotational-input-unit adapter **30**, to the rotational input unit of the bowstring drawing mechanism of the crossbow, causing the bowstring tension to be released in a controlled and safe manner.

[0050] Many other changes and modifications can be made in the system and method of the present invention without departing from the spirit thereof. I therefore pray that my rights to the present invention be limited only by the scope of the appended claims.

Claims

1. A rotational force assistance for crossbow drawing device for use with a crossbow having a bowstring drawing mechanism having a rotational input unit, the rotational force assistance for crossbow drawing device comprising: (i) a unit case adapted for portability and for single-hand use; (ii) a drive unit adapted to transform supplied electric power into rotational force; (iii) at least one rotational-input-unit adapter adapted to transfer rotational force from said drive unit to the rotational input unit of the bowstring drawing mechanism of the crossbow; (iv) a charge-storage unit adapted to store and release electric power for operation of said drive unit; (v) a charging port adapted to receive electric power for storage in said charge-storage unit; (vi) a draw activator adapted to activate said drive unit to apply rotational force in a tightening direction; and (vii) a controller adapted to control operation of said drive unit and said charge-storage unit.
2. The rotational force assistance for crossbow drawing device of claim 1, further comprising a bracing extension adapted for bracing against the crossbow structure to resist counter-rotational force.
3. The rotational force assistance for crossbow drawing device of claim 1, further comprising a retractable bracing extension adapted for bracing against the crossbow structure to resist counter-rotational force.
4. The rotational force assistance for crossbow drawing device of claim 1, further comprising a grip surface on said unit case.
5. The rotational force assistance for crossbow drawing device of claim 1, further comprising an indicator on said unit case, adapted to show status information.
6. The rotational force assistance for crossbow drawing device of claim 1, further comprising a tether point on said unit case, adapted to provide a mounting point for a tether.
7. The rotational force assistance for crossbow drawing device of claim 1, further comprising a release activator adapted to activate said drive unit to apply rotational force in a releasing direction.
8. The rotational force assistance for crossbow drawing device of claim 1, where said at least one rotational-input-unit adapters further comprise a plurality of rotational-input-unit adapters, each adapted to mount to the rotational input unit of the bowstring drawing mechanism of a specific crossbow.
9. The rotational force assistance for crossbow drawing device of claim 1, where said charging port further comprises a USB charging port.
10. The rotational force assistance for crossbow drawing device of claim 1, where said charge-storage unit further comprises a battery.
11. The rotational force assistance for crossbow drawing device of claim 1, where said charge-storage unit further comprises a capacitor.
12. A rotational force assistance for crossbow drawing device for use with a crossbow having a bowstring drawing mechanism having a rotational input unit, the rotational force assistance for crossbow drawing device comprising: (i) a unit case having a grip surface, adapted for portability and for single-hand use; (ii) a drive unit adapted to transform supplied electric power into rotational force; (iii) a plurality of rotational-input-unit adapters, each adapted to mount to the rotational input unit of the bowstring drawing mechanism of a specific crossbow, adapted to transfer rotational force from said drive unit to the rotational input unit of the bowstring drawing mechanism of the crossbow; (iv) a charge-storage unit adapted to store and release electric power for operation of said drive unit; (v) a charging port adapted to receive electric power for storage in said charge-storage unit; (vi) a draw activator adapted to activate said drive unit to apply rotational force in a

tightening direction; (vii) a release activator adapted to activate said drive unit to apply rotational force in a releasing direction; (viii) a controller adapted to control operation of said drive unit and said charge-storage unit; (ix) an indicator on said unit case, adapted to show status information; and (x) a tether point on said unit case, adapted to provide a mounting point for a tether.

13. The rotational force assistance for crossbow drawing device of claim 12, where said bracing extension further comprises a retractable bracing extension.

14. The rotational force assistance for crossbow drawing device of claim 12, where said charging port further comprises a USB charging port.

15. The rotational force assistance for crossbow drawing device of claim 12, where said charge-storage unit further comprises a battery.

16. The rotational force assistance for crossbow drawing device of claim 12, where said charge-storage unit further comprises a capacitor.

17. A rotational force assistance for crossbow drawing method for use with a crossbow having a bowstring drawing mechanism having a rotational input unit, the rotational force assistance for crossbow drawing method comprising: (i) providing a rotational force assistance for crossbow drawing device comprising: (a) a unit case adapted for portability and for single-hand use; (b) a drive unit adapted to transform supplied electric power into rotational force; (c) at least one rotational-input-unit adapter adapted to transfer rotational force from said drive unit to the rotational input unit of the bowstring drawing mechanism of the crossbow; (d) a charge-storage unit adapted to store and release electric power for operation of said drive unit; (e) a charging port adapted to receive electric power for storage in said charge-storage unit; (f) a draw activator adapted to activate said drive unit to apply rotational force in a tightening direction; and (g) a controller adapted to control operation of said drive unit and said charge-storage unit; (ii) mounting said rotational-input-unit adapter to said drive unit; (iii) mounting said rotational-input-unit adapter mounted to said drive unit to the rotational input unit of the bowstring drawing mechanism of the crossbow; (iv) engaging said draw activator; and (v) causing said rotational-input-unit adapter mounted to said drive unit to apply rotational force in a tightening direction to the rotational input unit of the bowstring drawing mechanism of the crossbow.

18. The rotational force assistance for crossbow drawing method of claim 17, further comprising providing said rotational force assistance for crossbow drawing device further comprising a release activator adapted to activate said drive unit to apply rotational force in a releasing direction; engaging said release activator; and causing said rotational-input-unit adapter mounted to said drive unit to apply rotational force in a releasing direction to the rotational input unit of the bowstring drawing mechanism of the crossbow.
